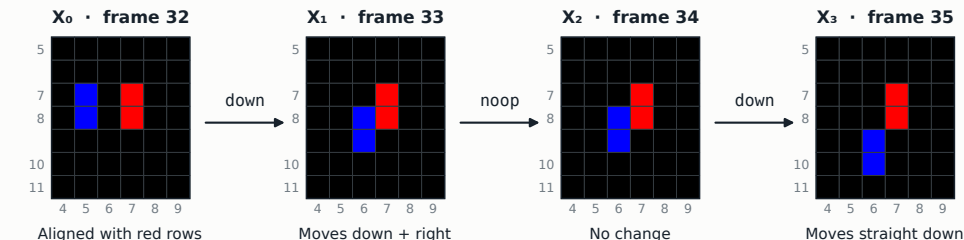


Game sequence →



Search checkpoints ↓

Iteration 0

P and B are empty

$P(X_0) = ""$

$P(X_1) = ""$

$P(X_2) = ""$

$P(X_3) = ""$

World knowledge B = $\emptyset$ · no learned rules

Iteration 4

P describes cells; B learns coordinate rules

$P(X_0)$
bg=black; cells:
7,5,blue
7,7,red
8,5,blue
8,7,red

$P(X_1)$
bg=black; cells:
7,7,red
8,6,blue
8,7,red
9,6,blue

$P(X_2)$
bg=black; cells:
7,7,red
8,6,blue
8,7,red
9,6,blue

$P(X_3)$
bg=black; cells:
7,7,red
8,7,red
9,6,blue
10,6,blue

World knowledge B — newly learned coordinate rules (excerpts)

... A pair of blue cells exists. ... the top cell's coordinates (r, c). ...
... down: (r, c) -> (r+1, c); if **c == 5** then additionally **c -> c+1**;
if **c == 9** then additionally **c -> c-1**. ...
... noop: **no change**. ...

Iteration 8

P adds relations; B uses those relations

rDistN: distance to red in the same row. **noRedRow**: no red in that row.

$P(X_0)$
7,5,blue, **rDist2**;
7,7,red;
8,5,blue, **rDist2**;
8,7,red

$P(X_1)$
7,7,red;
8,6,blue, **rDist1**;
8,7,red;
9,6,blue, **noRedRow**

$P(X_2)$
7,7,red;
8,6,blue, **rDist1**;
8,7,red;
9,6,blue, **noRedRow**

$P(X_3)$
7,7,red;
8,7,red;
9,6,blue, **noRedRow**;
10,6,blue, **noRedRow**

World knowledge B — rules using P's labels (excerpts)

... Vertical actions (up/down): ... For blue cells with label "**rDist**":
If there is a "**noRedRow**" blue in the **same column**, then:
The **rDist** blue **moves 1 row toward** that **noRedRow** blue ...
The **noRedRow** blue **moves 1 row further** in the same vertical direction.
Else (no "**noRedRow**" in the **same column**): ... For action "down":
The **rDist** blue in row 7 ... **moves down 2 rows to row 9**,
also **moves horizontally 1 step toward column 7**, and becomes "**noRedRow**".
The **rDist** blue in row 8 ... **moves horizontally 1 step toward column 7** ...
... The action "noop" **changes no feature**. ...